

sebastien fregonese

其他

基本信息

姓名: sebastien fregonese

所属院校: 波尔多大学

地区: France

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

射频集成电路

EDA工具与电路验证

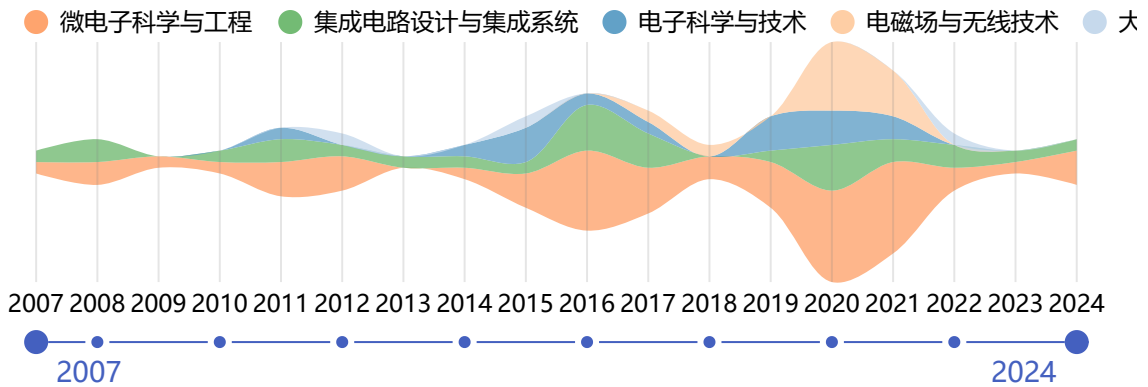
纳米电子与先进工艺

模拟与射频IC设计

个人简介

Sebastien Fregonese is a CR CNRS at the Université de Bordeaux. His email at ims-bordeaux.fr is verified. From 2020 to the present, he has 1210 citations, an h-index of 16, and an i10-index of 35, with a total of 2894 citations, an h-index of 25, and an i10-index of 80. He has 24 open-access articles. He has co-authored works with many researchers, including Thomas Zimmer, Marina Deng, and Anjan Chakravorty. His research is mainly in the field of electronics, with publications in journals like IEEE Transactions on Circuits and Systems I: Regular Papers and IEEE Transactions on Nanotechnology.

近年发表的论文



Next Generation SiGe HBTs for Energy Efficient Microwave Power Amplification (Invited)

8TH IEEE ELECTRON DEVICES TECHNOLOGY & MANUFACTURING CONFERENCE, EDTM 2024

Panda, Soumya Ranjan; Billy, Philippine; Gauthier, Alexis; Guitard, Nicolas; Chevalier, Pascal; De Matos, Magali; Zimmer, Thomas; Fregonese, Sebastien

被引用次数: 0 2024

Establishing On-Wafer Calibration Standards for the 16-Term Error Model: Application to Silicon High-Frequency Transistor Characterization

IEEE JOURNAL OF MICROWAVES

Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 0 2024

Exploring Compact Modeling of SiGe HBTs in Sub-THz Range With HICUM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Panda, Soumya Ranjan; Zimmer, Thomas; Chakravorty, Anjan; Derrier, Nicolas; Fregonese, Sebastien

被引用次数: 0 2024

A TCAD-Based Analysis of Substrate Bias Effect on Asymmetric Lateral SiGe HBT for THz Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Panda, Soumya Ranjan; Fregonese, Sebastien; Chevalier, Pascal; Chakravorty, Anjan; Zimmer, Thomas

被引用次数: 2 2023

Optimizing Finger Spacing in Multifinger Bipolar Transistors for Minimal Electrothermal Coupling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gupta, Aakashdeep; Nidhin, K.; Balanethiram, Suresh; Yadav, Shon; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

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A Technique for the In-Situ Experimental Extraction of the Thermal Impedance of Power Devices

IEEE TRANSACTIONS ON POWER ELECTRONICS

Scognamillo, Ciro; Fregonese, Sebastien; Zimmer, Thomas; d'Alessandro, Vincenzo; Catalano, Antonio Pio

被引用次数: 7 2022

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IEEE TRANSACTIONS ON ELECTRON DEVICES

Nidhin, K.; Balanethiram, Suresh; Nair, Deleep R. R.; D'Esposito, Rosario; Mohapatra, Nihar R. R.; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

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2022

Reliable Technology Evaluation of SiGe HBTs and MOSFETs: fMAX Estimation From Measured Data

IEEE ELECTRON DEVICE LETTERS

Saha, Bishwadeep; Fregonese, Sebastien; Heinemann, Bernd; Scheer, Patrick; Chevalier, Pascal; Aufinger, Klaus; Chakravorty, Anjan; Zimmer, Thomas

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2021

Extraction of True Finger Temperature From Measured Data in Multifinger Bipolar Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gupta, Aakashdeep; Nidhin, K.; Balanethiram, Suresh; D'Esposito, Rosario; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

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2021

Importance of Probe Choice for Extracting Figures of Merit of Advanced mmW Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; De Matos, Magali; Deng, Marina; Celi, Didier; Derrier, Nicolas; Zimmer, Thomas

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2021

Meander-Type Lines: An Innovative Design for On-Wafer TRL Calibration for mmW and Sub-mmW Frequencies Measurements

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Cabbia, Marco; Fregonese, Sebastien; Deng, Marina; Curutchet, Arnaud; Yadav, Chandan; Celi, Didier; De Matos, Magali; Zimmer, Thomas

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2021

Investigation of Variation in On-Si On-Wafer TRL Calibration in Sub-THz

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Yadav, Chandan; Deng, Marina; Fregonese, Sebastien; Cabbia, Marco; De Matos, Magali; Zimmer, Thomas

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2021

Guideline for Test-Structures Placement for on-Wafer Calibration in sub-THz Si Device Characterization

2021 IEEE MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM (IMS)

Yadav, Chandan; Cabbia, Marco; Fregonese, Sebastien; Deng, Marina; De Matos, Magali; Zimmer, Thomas

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2021

Electro-thermal limitations and device degradation of SiGe HBTs with emphasis on circuit performance

2021 IEEE BICMOS AND COMPOUND SEMICONDUCTOR INTEGRATED CIRCUITS AND TECHNOLOGY SYMPOSIUM (BCICTS)

Fregonese, Sebastien; Mukherjee, Chhandak; Ruecker, Holger; Chevalier, Pascal; Fischer, Gerhard; Celi, Didier; Deng, Marina; Couret, Marine; Marc, Francois; Maneux, Cristell; Zimmer, Thomas

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Performance prediction of InP/GaAsSb double heterojunction bipolar transistors for THz applications

JOURNAL OF APPLIED PHYSICS

Wen, Xin; Arabhavi, Akshay; Quan, Wei; Ostinelli, Olivier; Mukherjee, Chhandak; Deng, Marina; Fregonese, Sebastien; Zimmer, Thomas; Maneux, Cristell; Bolognesi, Colombo R.; Luisier, Mathieu

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TCAD and EM co-simulation method to verify SiGe HBT measurements up to 500 GHz

SOLID-STATE ELECTRONICS

Panda, Soumya Ranjan; Fregonese, Sebastien; Deng, Marina; Chakravorty, Anjan; Zimmer, Thomas

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An Efficient Thermal Model for Multifinger SiGe HBTs Under Real Operating Condition

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2020

Analysis of High-Frequency Measurement of Transistors Along With Electromagnetic and SPICE Cosimulation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; Cabbia, Marco; Yadav, Chandan; Deng, Marina; Panda, Soumya Ranjan; De Matos, Magali; Celi, Didier; Chakravorty, Anjan; Zimmer, Thomas

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2020

THz Characterization and Modeling of SiGe HBTs: Review (Invited)

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Fregonese, Sebastien; Deng, Marina; Cabbia, Marco; Yadav, Chandan; De Matos, Magali; Zimmer, Thomas

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Silicon Test Structures Design for Sub-THz and THz Measurements

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2020

Design of On-Wafer TRL Calibration Kit for InP Technologies Characterization up to 500 GHz

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Static Thermal Coupling Factors in Multi-Finger Bipolar Transistors: Part I-Model Development

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Importance and Requirement of Frequency Band Specific RF Probes EM Models in Sub-THz and THz Measurements up to 500 GHz

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Yadav, Chandan; Deng, Marina; Fregonese, Sebastien; Cabbia, Marco; De Matos, Magali; Plano, Bernard; Zimmer, Thomas

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High-Frequency Noise Characterization and Modeling of Graphene Field-Effect Transistors

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

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Comparison of On-Wafer TRL Calibration to ISS SOLT Calibration With Open-Short De-Embedding up to 500 GHz

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2019

A Multiscale TCAD Approach for the Simulation of InP DHBTs and the Extraction of Their Transit Times

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Validation of Thermal Resistance Extracted From Measurements on Stripe Geometry SiGe HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

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2019

Scalable Modeling of Thermal Impedance in InP DHBTs Targeting Terahertz Applications

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Scalable Compact Modeling of III-V DHBTs: Prospective Figures of Merit Toward Terahertz Operation

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A Large-Signal Monolayer Graphene Field-Effect Transistor Compact Model for RF-Circuit Applications

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2017

Accurate Modeling of Thermal Resistance for On-Wafer SiGe HBTs Using Average Thermal Conductivity

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Extraction of BEOL Contributions for Thermal Resistance in SiGe HBTs

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Effects of BEOL on self-heating and thermal coupling in SiGe multi-finger HBTs under real operating condition

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On the development of a novel high VSWR programmable impedance tuner

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Curutchet, Arnaud; Ghiotto, Anthony; Potereau, Manuel; De Matos, Magali; Fregonese, Sebastien; Kerherve, Eric; Zimmer, Thomas

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A Study on Self-Heating and Mutual Thermal Coupling in SiGe Multi-Finger HBTs

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Dwivedi, A. D. D.; D'Esposito, Rosario; Sahoo, Amit Kumar; Fregonese, Sebastien; Zimmer, Thomas

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Efficient Modeling of Distributed Dynamic Self-Heating and Thermal Coupling in Multifinger SiGe HBTs

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Innovative SiGe HBT Topologies With Improved Electrothermal Behavior

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D'Esposito, Rosario; Fregonese, Sebastien; Chakravorty, Anjan; Chevalier, Pascal; Celi, Didier; Zimmer, Thomas

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2016

Comments on Optimization of a Compact I-V Model for Graphene FETs: Extending Parameter Scalability for Circuit Design Exploration

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Fregonese, Sebastien; Zimmer, Thomas

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Graphene Transistor-Based Active Balun Architectures

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zimmer, Thomas; Fregonese, Sebastien

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A Geometry Scalable Model for Nonlinear Thermal Impedance of Trench Isolated HBTs

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2015

Electrical Compact Modeling of Graphene Base Transistors

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An Accurate Physics-Based Compact Model for Dual-Gate Bilayer Graphene FETs

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Innovative Dual-Gate CNTFET Logic Cell: Investigation of Technological Dispersion Impact Through Compact Modeling

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A Comprehensive Graphene FET Model for Circuit Design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rodriguez, Saul; Vaziri, Sam; Smith, Anderson; Fregonese, Sebastien; Ostling, Mikael; Lemme, Max C.; Rusu, Ana

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Scalable Electrical Compact Modeling for Graphene FET Transistors

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2012

Design and Modeling of a Neuro-Inspired Learning Circuit Using Nanotube-Based Memory Devices

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Liao, Si-Yu; Retrouvey, Jean-Marie; Agnus, Guillaume; Zhao, Weisheng; Maneux, Cristell; Fregonese, Sebastien; Zimmer, Thomas; Chabi, Djaafar; Filoramo, Arianna; Derycke, Vincent; Gamrat, Christian; Klein, Jacques-Olivier

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Schottky Barrier Carbon Nanotube Transistor: Compact Modeling, Scaling Study, and Circuit Design Applications

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A Compact Model for Dual-Gate One-Dimensional FET: Application to Carbon-Nanotube FETs

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Computationally efficient physics-based compact CNTFET model for circuit design

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Fregonese, Sebastien; d'Honinckhun, Hugues Cazin; Goguet, Johnny; Maneux, Cristell; Zimmer, Thomas; Bourgoin, Jean-Philippe; Dollfus, Philippe; Galdin-Retailleau, Sylvie

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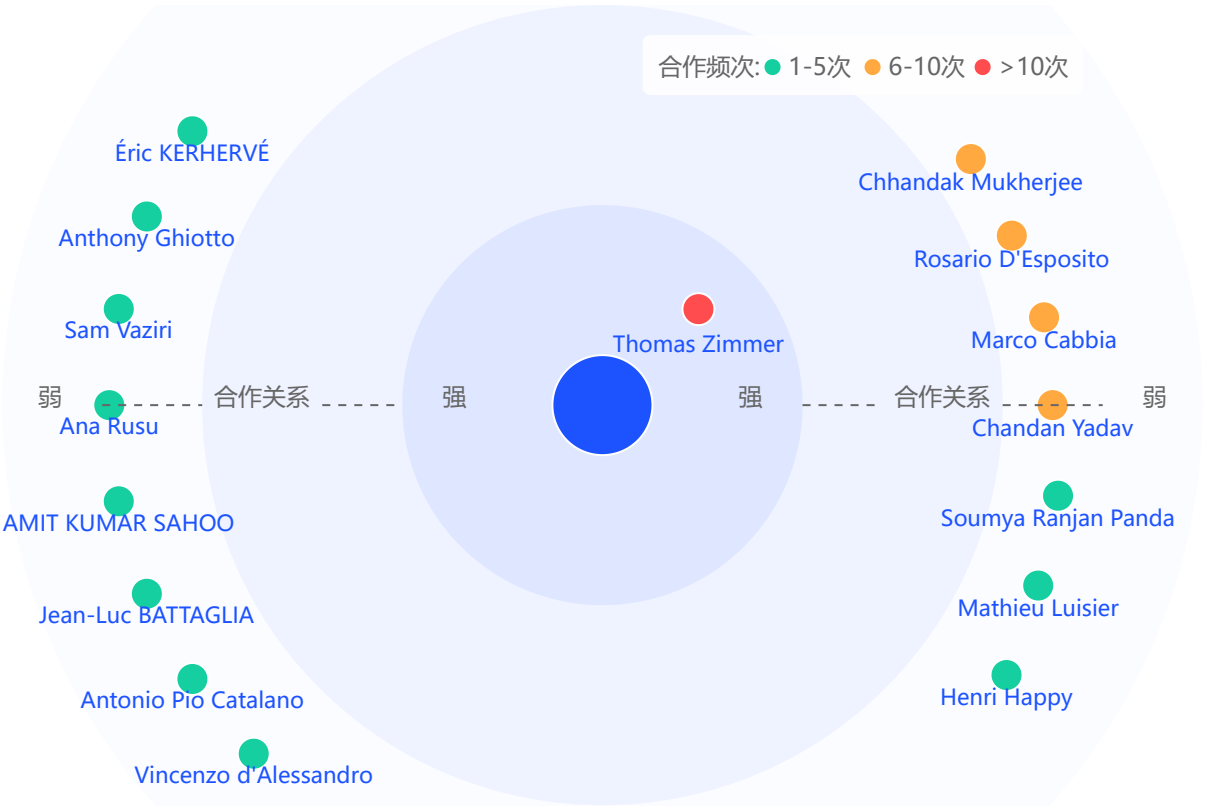
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2008

学术合作



Thomas Zimmer	合作次数：51
波尔多大学	
Chhandak Mukherjee	合作次数：8
波尔多大学	
Rosario D'Esposito	合作次数：8
波尔多大学	
Marco Cabbia	合作次数：7
波尔多大学	
Chandan Yadav	合作次数：7
波尔多大学	

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